



Total RNA Plant Viriome Analysis Report

TCP_2627_06

Dr.Mahesh

IIHR

23-05-2026

Table of Contents

- **RNA Library Prep Protocol**
- **Executive Data Analysis Summary**
- **Methods – Data Analysis**
- **Bioinformatics Pipeline**
- **Results**
- **Deliverables**

RNA Library Prep Protocol

Total RNA was extracted from 30–50 mg of leaf tissue using TRIzol Reagent (Thermo Fisher Scientific). Samples were homogenized and incubated for 5 minutes to dissociate nucleoprotein complexes. Phase separation was achieved by adding 0.2 mL of chloroform per 1 mL of TRIzol, followed by centrifugation at maximum speed. The aqueous phase was transferred to a clean tube, and RNA was precipitated using 0.5 mL of isopropanol. The resulting pellet was washed with 75% ethanol, air-dried for 5–10 minutes, and resuspended in 25 µL of RNase-free water. Samples are quantified using the Agilent BioAnalyzer 2100 and Invitrogen Qubit 4 HS RNA assay(Q32855)

NEB Ultra II FS DNA lib prep protocol

rRNA depletion using the Qiagen HMR (334387). To perform depletion, 500ng of total RNA is mixed with FastSelect reagent and denaturation buffer, then incubated through a short thermal protocol (e.g., 70°C for 5 minutes, then ramping down to 25°C) to allow hybridization of FastSelect oligos to rRNA targets. After rRNA depletion, total RNA prepared using NEBNext® Ultra™ II Directional RNA Library Prep Kit for Illumina(E7760L). RNA fragmentation and priming are carried out by incubating the RNA with NEBNext First Strand Synthesis Buffer and Random Primers at 94°C for 8 minutes. First strand cDNA synthesis follows using reverse transcriptase at 25°C for 10 minutes, 42°C for 15 minutes, and 70°C for 15 minutes. Second strand synthesis incorporates dUTP for strand specificity and proceeds at 16°C for 1 hour, followed by cleanup with AMPure XP beads. End repair and dA-tailing are performed by incubating with End Prep reagents at 20°C for 30 minutes and 65°C for 30 minutes. Adapter ligation is carried out at 20°C for 15 minutes, followed by USER enzyme treatment at 37°C for 15 minutes. After the bead purification, PCR enrichment (8 cycles) is performed with NEBNext Q5 Master Mix and indexed primers. The final library is cleaned, assessed for fragment size using the Agilent TapeStation 4150 and Invitrogen Qubit 4 HS DNA assay(Q32851)Quantified libraries sequenced using Illumina Novaseq 6000.

Executive Data Analysis Summary

A single plant sample was analyzed using a total RNA sequencing-based virome detection pipeline for the identification and characterization of known RNA viruses. Raw sequencing reads underwent quality assessment, adapter trimming, and ribosomal RNA (rRNA) removal to enrich for non-rRNA sequences. The resulting reads were subjected to host subtraction by mapping against the mango reference genome (*Selenicereus_undatus*; GCA_017589665.1) to remove host-derived reads. Host-subtracted reads were assembled de novo into contigs, which were subsequently filtered based on sequence length and screened against viral reference databases using BLASTN. Significant viral matches were retained following predefined filtering criteria and considered candidate viral contigs. Sequencing reads were then remapped to the BLASTN-identified viral regions to quantify viral read abundance and estimate the relative abundance of detected viruses. To support functional characterization, only the high-confidence viral contigs identified through BLASTN analysis were subjected to protein-level homology searches using DIAMOND BLASTX against a curated viral protein database. Conserved viral proteins and functional domains identified through BLASTX were used to further support the taxonomic assignment of the detected viruses. Coverage validation was subsequently performed exclusively on the filtered BLASTN-supported viral contigs by remapping sequencing reads, thereby providing additional evidence for the presence and completeness of the identified viral sequences.

Methods – Data Analysis

Bioinformatics Pipeline

The following bioinformatics steps were performed for analysis of the data

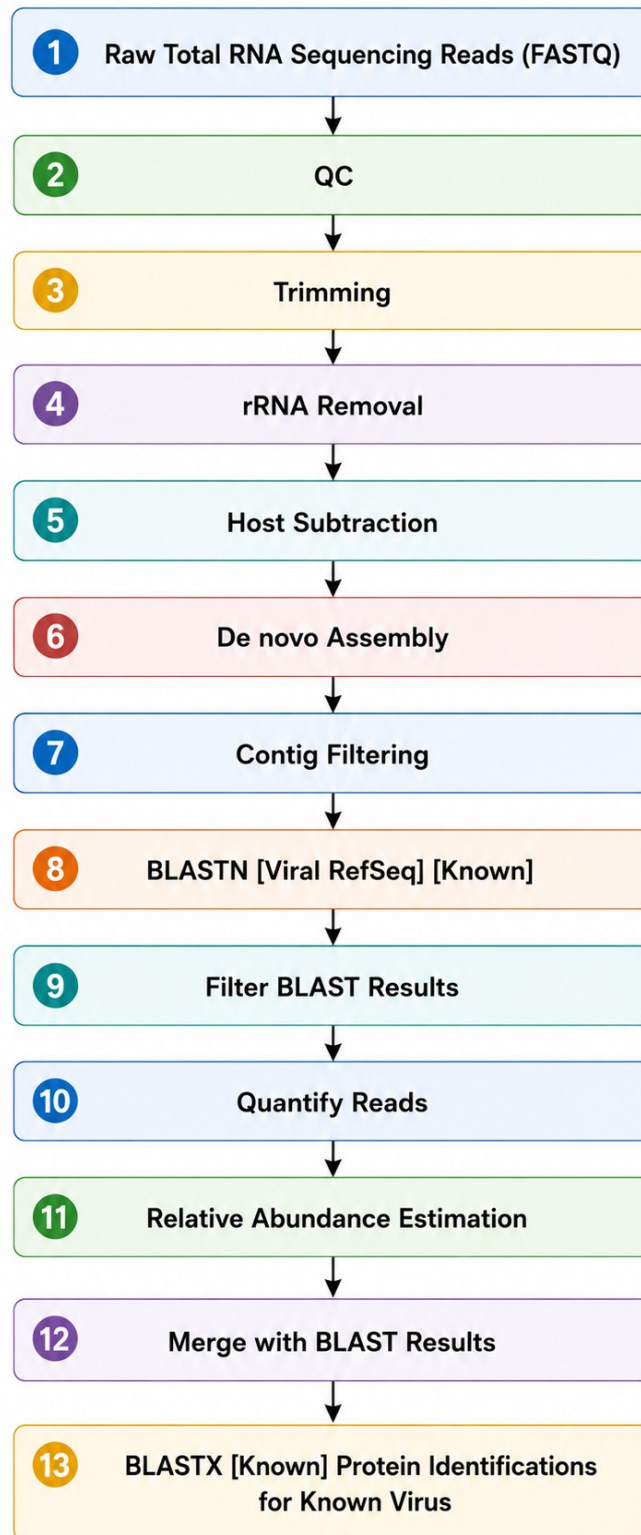


Fig 1. Bioinformatics workflow for RNA virus Plant virome analysis

Bioinformatic Steps

Read quality check

The following parameters from fastq file were checked as part of quality check- Base quality score distribution, Sequence quality score distribution, Average base content per read, GC distribution in the reads, PCR amplification issue, Check for over-represented sequences and Adapter trimming.

Adapter Trimming

Based on the quality report of fastq files trimming of raw read was performed where necessary to only retain high quality sequence for further analysis. In addition, the low-quality sequence reads are excluded from the analysis. The adapter trimming was performed using Cutadapt (v-3.5).

rRNA Removal

To enrich for viral RNA sequences, ribosomal RNA (rRNA) reads were removed from the trimmed dataset using SortMeRNA v4.3.4. Reads were screened against the SortMeRNA v4.3.4 database bundle, which incorporates SILVA SSU (16S/18S), SILVA LSU (23S/28S), RFAM 5S, and RFAM 5.8S rRNA reference sequences. Reads classified as rRNA were excluded, while the remaining non-rRNA reads were retained for host subtraction and virome analysis.

Host Subtraction

Non-rRNA reads were aligned against the mango reference genome *Selenicereus_undatus* (NCBI RefSeq assembly GCA_017589665.1) using Bowtie2. Reads mapping to the host genome were removed, and only unmapped reads were retained for downstream viral discovery. This host subtraction step reduces interference from host-derived transcripts and improves the sensitivity of viral sequence detection.

De Novo Assembly

Host-subtracted reads were assembled de novo using rnaSPAdes with a multi-k-mer assembly strategy (k = 21, 33, 55, and 77). To maximize sensitivity for low-abundance viral sequences, coverage-based contig filtering during assembly was disabled (--cov-cutoff off). Assembly statistics, including total assembly size, number of contigs, N50, and largest contig length, were subsequently evaluated using QUAST.

Contig Filtering

To reduce the number of low-complexity and potentially artifactual assemblies, assembled contigs were filtered based on length using Seqtk. A minimum contig length threshold of 200 bp was applied to retain potential viral sequences while excluding short assembly fragments. The filtered contig set was used for all downstream viral identification analyses.

Viral Identification by BLASTN

Filtered contigs were screened against the NCBI Viral RefSeq nucleotide database(viral.1.1.genomic.fna) using BLASTN to identify known viral sequences. Searches were performed using an E-value threshold of 1×10^{-10} , a minimum nucleotide identity of 80%, minimum query coverage per high-scoring segment pair(qcovhsp) of 80%, and a maximum of one high-scoring segment pair (HSP) per query sequence. Additional searches were performed against a plant-virus-focused nucleotide database (Virosaurus90p 2020_4.1) to improve detection sensitivity for plant-associated viruses and viroids. BLASTN results were subsequently filtered to retain the highest-confidence viral matches.

BLASTN Result Filtering

BLASTN results were filtered to generate a high-confidence set of viral contigs for downstream quantification and validation. For contigs with a single viral hit, the alignment was retained without modification. When multiple hits occurred on the same contig with identical query start and query end coordinates, only the hit with the highest query coverage per high-scoring segment pair (qcovhsp) was retained. For contigs containing multiple overlapping viral hits with different alignment coordinates, the hit spanning the largest query region, defined by the minimum query start and maximum query end positions among the overlapping alignments, was retained as the representative viral match. Hits occurring on distinct, non-overlapping regions of the same contig were considered independent viral detections and retained separately. This filtering strategy minimized redundant annotations while preserving evidence for potentially distinct viral sequences present within a single assembled contig.

Viral Quantification and Relative Abundance Estimation

Reads were remapped to the filtered viral regions using Bowtie2 in very-sensitive local alignment mode. Viral regions identified through BLASTN analysis were converted into Simplified Annotation Format (SAF) files and quantified using featureCounts. The resulting read counts were used to estimate the abundance of each detected virus and to calculate relative abundance values within the sample.

Protein-Level Functional Characterization

High-confidence viral contigs identified through BLASTN analysis were further characterized using DIAMOND BLASTX against the RVDB protein database (U-RVDB v31.0). Searches were performed using an E-value threshold of 1×10^{-10} , minimum amino acid identity of 80% and a maximum of one HSP per query. Protein-level homology searches were used to identify conserved viral proteins and functional domains, including RNA-dependent RNA polymerases, coat proteins, movement proteins, helicases, and proteases, thereby providing additional support for viral classification.

Coverage Validation

To validate the presence of identified viruses, sequencing reads were remapped to the filtered viral contigs that satisfied the BLASTN filtering criteria. Coverage statistics were calculated using SAMtools, including mean sequencing depth and breadth of genome coverage. Coverage validation provided independent evidence supporting the authenticity, completeness, and abundance of the detected viral sequences.

Samples and Grouping information

Following samples were sequenced and analyzed.

Table 1. Samples Information

Sl.No	theraCUES Sample ID	Client Sample ID	Sample.Type
1	SAM8483	MD1	Dragon Fruit

Results

Data QC Summary

Table 3 summarizes the overall data generated and the average read quality observed across the reads.

Table 3. Raw Data QC Summary

Client Sample ID	Sample .Type	Total paired reads (Before adapter trimming)	Data generated (Gb)	Total paired reads (After adapter trimming)	Percentage of reads remaining after trimming (%)	Avg.base quality(phred)	Total data (reads) $\geq$ Q30 %	GC %
MD1	Dragon fruit	9139546	2.76	9107735	99.65	39.3	98.70	47

Alignment Summary

Alignment of quality-filtered reads against the mango reference genome resulted in an overall alignment rate of 74.20%, indicating that the majority of sequencing reads were host-derived.

De Novo Assembly and Contig Filtering

Host-subtracted reads were assembled de novo, generating a total of 15,113 contigs. The assembled contigs ranged up to a maximum length of 7,917 bp and exhibited an overall GC content of 44.28%. Assembly quality metrics showed an N50 of 694 bp, an N90 of 531 bp, and an auN value of 1,000.8 bp. To reduce the inclusion of short assembly fragments and potential assembly artifacts, contigs were filtered using a minimum length threshold of 200 bp.

Viral Identification by BLASTN

Filtered contigs were screened against viral nucleotide reference databases using BLASTN. Analysis against the NCBI Viral RefSeq nucleotide database identified 12 viral contigs with significant sequence similarity to known viral genomes. To improve the detection of plant-associated viruses, the contigs were also screened against a plant-specific viral database, resulting in the identification of 7 viral contigs. The increased number of viral detections obtained using the plant-focused database highlights the benefit of utilizing host-relevant viral reference collections for plant virome studies.

Viral Quantification and Relative Abundance

Filtered viral regions were quantified by remapping sequencing reads and performing read counting using featureCounts. Based on the Viral RefSeq nucleotide database results, *Pitaya virus X isolate P37* was identified as the most abundant viral sequence detected in the sample. Similarly, analysis using the plant-specific viral database identified *Cactus virus X* as the most abundant virus. The plant-specific database provided improved representation of plant-associated viruses and closely related cactus-infecting viral sequences, thereby enabling more refined taxonomic assignment of the detected viral contigs.

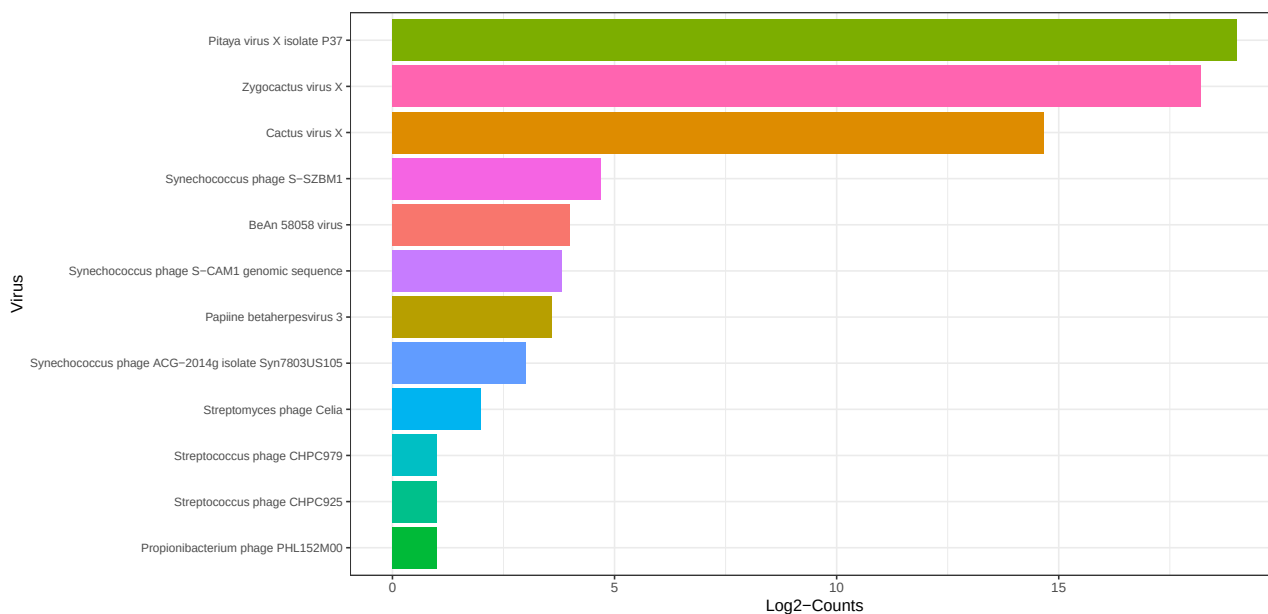


Fig 2. Bar plot representing relative abundance of viruses using viral Refseq nucleotide databases

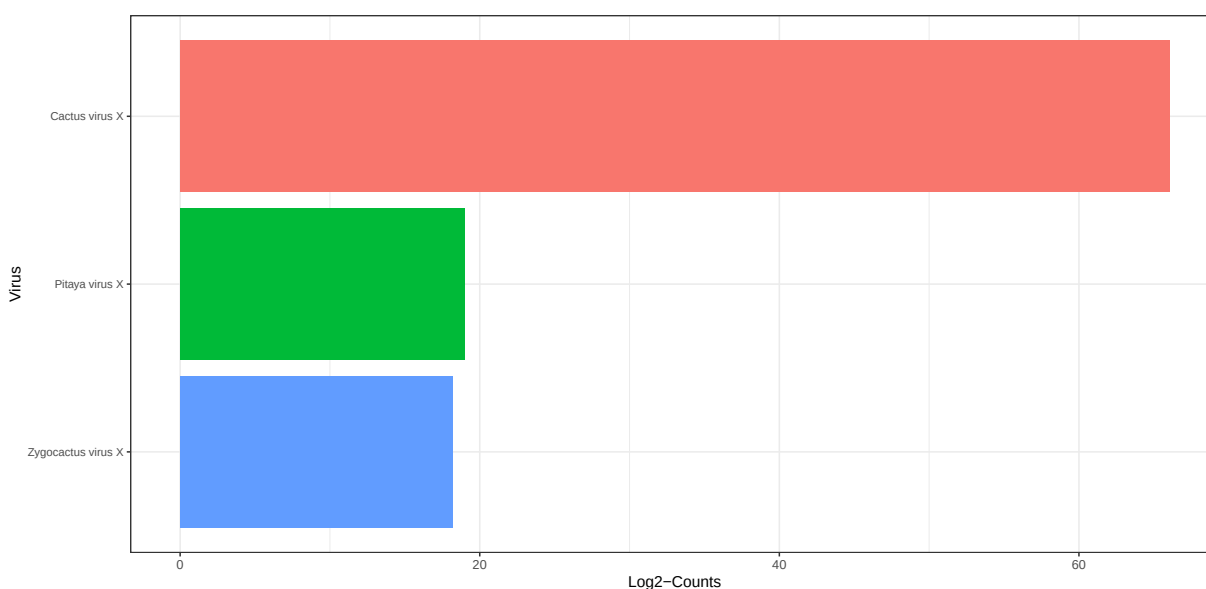


Fig 3. Bar plot representing relative abundance of viruses using databases plant-virus-focused nucleotide database

Protein-Level Functional Characterization

The high-confidence BLASTN-supported viral contigs were subjected to DIAMOND BLASTX analysis against the RVDB protein database. Conserved viral proteins including RNA-dependent RNA polymerase (RdRp) were identified, providing additional support for the taxonomic classification of the detected viruses.

Coverage Validation

Coverage validation demonstrated complete breadth coverage (100%) across all 12 high-confidence viral contigs. Mean sequencing depth ranged from 1.19× to 13,609.8×, with several contigs exhibiting exceptionally high read support (>7,000× depth). The observed coverage profiles provide strong evidence supporting the authenticity and abundance of the detected viral sequences.

Final Deliverables

1. Quality Control Report
2. Host-subtracted (unmapped) reads retained for analysis
3. De Novo Assembly Report
4. BLASTN Annotation Report
5. BLASTX (DIAMOND) Annotation Report
6. Viral Quantification Report
7. Coverage Validation Report
8. Virome Analysis Summary Report

Access to all reports :

<https://workdrive.zohoexternal.com/external/38a544d676588ea8b56105ffc66d07a5127e02b13ebe71d954860ac0a3ea66c4>
